

SONALI VYAS

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[sonalivyas/cse19-stack.github.io](https://github.com/sonalivyas/cse19-stack) | github.com/sonalivyas/cse19-stack

PROFESSIONAL SUMMARY

AI Engineer and Software Engineer with 6 years of experience building and shipping production systems across Python, Java, and cloud platforms. Deep expertise in Salesforce ecosystem engineering, enterprise system integration, and event-driven microservices architecture. Currently conducting machine learning and data assimilation research at the University of Alabama in collaboration with NOAA. Experienced in designing multi-agent AI systems, deploying LLM-powered applications, and connecting complex technical systems with real-world business outcomes. Comfortable leading backend architecture, shipping full-stack applications, or designing agentic AI pipelines from scratch.

TECHNICAL SKILLS

AI and ML: LLM integration, Multi-Agent Systems, LangGraph, CrewAI, PyTorch, data assimilation, ML model deployment, prompt engineering, agentic pipeline design

Python Engineering: FastAPI, data pipelines, scientific computing, ML frameworks, REST API design, automation scripting

Java Engineering: Spring Boot, Apex (Salesforce), OOP design patterns, microservices, enterprise service integration

Cloud and Salesforce: Salesforce Sales Cloud, Experience Cloud, Apex, SOQL, Lightning Web Components, Salesforce Next Best Action, AWS EC2, Lambda, S3, Docker, Kubernetes

Integration and Architecture: REST APIs, event-driven microservices, cross-platform system integration, CI/CD via GitHub Actions

Frontend: React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Angular, Lightning Web Components

Databases: PostgreSQL, MySQL, SOQL

EDUCATION

M.S. in Computer Science | The University of Alabama

Expected May 2027

Tuscaloosa, Alabama | Coursework: Computer Algorithms, Artificial Intelligence, Computer Vision

B.Tech in Computer Science, Honors | Rajasthan Technical University

May 2019

Rajasthan, India

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | The University of Alabama

Jan 2026 – Present

Tuscaloosa, Alabama | Python, Machine Learning, Data Assimilation, Scientific Computing | Part-time

- Contributing to NOAA's Next Generation Water Resources Modeling Framework, a national-scale environmental simulation initiative.
- Developing machine learning and data assimilation capabilities to model hydrosphere responses at national scale, integrating real-world observational data with predictive ML pipelines.
- Applying scientific Python stack to environmental modeling problems requiring high reliability, reproducibility, and computational performance.

Senior Software Engineer II | American Express

Sep 2021 – Jul 2025

Bangalore, India | Python, Java, Apex, React, REST APIs, Microservices, Docker, Salesforce, GitHub Actions

Promoted three times: Engineer II (Sep 2021), Engineer I (Jul 2023), Senior Engineer II (Sep 2024)

- Architected and delivered an AI-powered case management system (OCMT) using Salesforce Next Best Action to auto-suggest resolution paths; designed a skill and availability-based agent assignment engine with real-time integration across multiple Amex platforms. Reduced cardmember wait times by 40% and saved agents 15 or more hours per week.
- Led full-stack rebuild of the MDT offer enrollment platform serving thousands of transactions per day, using event-driven microservices, React and Angular frontend, and Apex and SOQL backend. Reduced user clicks by 40% and improved page load speed by 30%.
- Built a Lightning Web Component UI enabling non-technical business users to configure granular merchant targeting criteria, driving approximately 7 million dollars in projected annual revenue.
- Designed secure integrations connecting Amex internal platforms with Salesforce, enabling real-time data flow across systems with strict compliance and reliability requirements.
- Set up and maintained end-to-end CI/CD pipelines via GitHub Actions covering automated testing, Docker builds, and production deployments across multiple projects.

Success Engineer | Salesforce Inc.

Jan 2021 – Sep 2021

Hyderabad, India

- Implemented Sales Cloud and Experience Cloud features for global enterprise clients; improved user adoption by 30% through customized configurations, stakeholder training, and technical support.
- Acted as a technical bridge between Salesforce platform capabilities and client business needs, translating complex requirements into working solutions.

Project Engineer | Wipro

May 2019 – Dec 2020

Jaipur, India

- Built an automated CPQ quoting system using JavaScript, HTML5, and CSS3 end-to-end; reduced partner portal quote generation time by 50% through streamlined configuration and automation.

PROJECTS

CrisisAssist AI | Personal Project

Aug 2025 – Present

Python, FastAPI, React, LangGraph, CrewAI, PyTorch, AWS, Docker, Kubernetes, GitHub Actions

- Designed and built a production-grade multi-agent AI system for emergency response support, with specialized agents coordinating across a real-time event detection pipeline.
- Full-stack application: React frontend, FastAPI backend, PyTorch ML model served via REST API, deployed on AWS with Docker and Kubernetes.
- Demonstrates applied expertise in agentic AI architecture, LLM orchestration, and reliable production deployment at scale.

AmexTechHub | American Express Hackathon, GrowthHack 2023 Semi-Finalist

Oct – Dec 2023

Python, Salesforce, Slack API, Confluence, AI and ML

- Built a knowledge management platform converting static internal documents into a searchable, AI-augmented system with integration across Salesforce, Slack, and Confluence.
- Recognized as a company-wide semi-finalist for technical execution and product quality.